



- 5 (a) State the relationship between electric field and electric potential.

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..... [2]

- (b) Two charged isolated insulating spheres X and Y are near to each other, as shown in Fig. 5.1.



Fig. 5.1

P is a point on the line joining the centres of the spheres.

Explain why it is **not** possible for the total electric potential and the resultant electric field to simultaneously be zero at point P.

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..... [3]

- (c) The magnitudes of the charges on spheres X and Y in Fig. 5.1 are Q and $2Q$ respectively. The spheres may be considered as point charges at their centres.

Point P is a distance x from the centre of sphere X.

The electric potential at point P is zero.

- (i) Show that the distance y of point P from the centre of sphere Y is equal to $2x$.

[2]





- (ii) State an expression, in terms of Q , x and the permittivity of free space ϵ_0 , for the electric field strength E_x at P due to sphere X.

$$E_x = \dots\dots\dots [1]$$

- (iii) Determine an expression, in terms of Q , x and ϵ_0 , for the resultant electric field strength E at point P due to the two spheres.

$$E = \dots\dots\dots [2]$$

[Total: 10]

